

NuraVolt

BESS Performance Health Checklist

Monitor SOC drift, temperature gradients, voltage imbalance & efficiency loss

Purpose: Use this checklist monthly (minimum) to detect early warning signs of battery degradation and thermal runaway risk. These indicators prevent warranty invalidation and catastrophic thermal events. Complete this checklist during routine operation and after any charge/discharge anomalies. Keep records for warranty claims and insurance compliance.

⚠ CRITICAL: Any parameter in "Critical" range requires immediate investigation and may necessitate module isolation to prevent thermal runaway and warranty invalidation.

1 Cell Voltage Imbalance



What to Check: Maximum voltage spread between highest and lowest cell within each module

Normal: <50 mV during operation

Warning: 50-100 mV or increasing >10 mV/week

Critical: >100 mV or >150 mV at rest

When to Check:

- During steady-state charge or discharge (0.5C rate preferred)
- After full equalization charge (100% SOC, held until balanced)
- Monthly trend analysis (plot max spread over time)

Action if Warning/Critical: Identify which cell(s) are outliers; check BMS sensor calibration; review module temperature distribution; inspect for signs of internal short circuit or dendrite formation; initiate warranty claim process if imbalance >150 mV and accelerating.

2 Temperature Gradient Across Cells



What to Check: Temperature difference between hottest cell and module average temperature

Normal: $<3^{\circ}\text{C}$
gradient

Warning: $3\text{-}5^{\circ}\text{C}$ sustained >15
min

Critical: $>5^{\circ}\text{C}$ or any cell
 $>45^{\circ}\text{C}$

When to Check:

- During peak charge/discharge operations (1C rate)
- Continuous monitoring during summer months (ambient $>40^{\circ}\text{C}$)
- After any BMS temperature alarm or HVAC fault

Action if Warning/Critical: Reduce charge/discharge rate immediately (derate to 0.5C); verify HVAC cooling system operational and unobstructed; check thermal paste on cell connections; consider module isolation if hotspot $>48^{\circ}\text{C}$; thermal imaging inspection to identify exact hot cell location; document for warranty claim if recurring issue.

3 State of Charge (SOC) Drift



What to Check: SOC difference between cells after full equalization cycle

Normal: $<2\%$ after
equalization

Warning: $2\text{-}5\%$ persistent
drift

Critical: $>5\%$ or cannot
equalize

When to Check:

- Weekly for NMC chemistry; monthly for LFP chemistry
- After equalization charge cycle (100% SOC hold for 2+ hours)
- Before any capacity testing or warranty claim

Action if Warning/Critical: Perform additional equalization cycles (may take 2-3 cycles for LFP); if drift persists $>5\%$ after 3 equalization attempts, indicates cell-level capacity loss or BMS Coulomb counting error; recalibrate BMS SOC algorithm; document for warranty claim as evidence of premature capacity degradation.

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State of Health (SOH) Degradation Rate



What to Check: Monthly capacity fade rate compared to expected degradation

Normal: 1-3%/yr (LFP), 2-5%/yr (NMC)

Warning: 1.5-2x expected rate

Critical: >2x expected or sudden 5% drop

When to Check:

- Monthly via full 0.2C discharge cycle (from 100% to 20% SOC)
- Calculate: $(\text{Current Capacity} / \text{Nameplate Capacity}) \times 100\%$
- Track trend over 12+ months to identify acceleration

Action if Warning/Critical: Review operating conditions: temperature excursions >45°C, C-rate violations >1C, SOC window violations; reduce charge/discharge rates by 30% to slow degradation; investigate warranty claim if SOH falls below manufacturer guarantee curve (typically 80% at 10 years or 10,000 cycles); document capacity tests with independent witness for warranty evidence.

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Round-Trip Efficiency Loss



What to Check: Energy efficiency = $(\text{Discharge Energy} / \text{Charge Energy}) \times 100\%$

Normal: 88-95% (chemistry dependent)

Warning: 85-88% or declining >3% in 6 months

Critical: <85% or rapid 5% drop

When to Check:

- Weekly calculation from SCADA energy meters
- Normalize for ambient temperature (efficiency decreases ~0.5%/10°C increase)
- Track monthly moving average to filter out daily noise

Action if Warning/Critical: Efficiency loss indicates internal resistance increase from cell degradation, connection resistance, or cooling system inadequacy; measure DC resistance of modules (should be <0.5 mΩ per cell); check busbar connection torque (thermal cycling loosens connections); review HVAC performance (high cell temperature increases losses); efficiency <85% may indicate accelerated warranty degradation.



What to Check: Time from power command to 90% power delivery

Normal: <500 ms
(typically <200 ms)

Warning: 500-1000 ms or
increasing trend

Critical: >1000 ms or power
oscillations

When to Check:

- During commissioning and quarterly thereafter
- After any firmware updates or BMS configuration changes
- When BESS fails to meet grid frequency response requirements

Action if Warning/Critical: Slow response indicates BMS control loop instability, contactor degradation, or DC link capacitor aging; review BMS control parameters and update firmware if available; test contactors for switching time and contact resistance; for grid services (frequency regulation), response >500ms may violate contractual requirements and result in revenue loss.



What to Check: Cell temperature vs ambient temperature during operation

Normal: Cell temp

Warning: Cell temp = ambient + 15-25°C

Critical: Cell temp > ambient + 25°C or >45°C absolute

When to Check:

- Daily during summer months (ambient >35°C)
- After HVAC maintenance or filter replacement
- When any cell temperature alarm occurs

Action if Warning/Critical: Check HVAC compressor operation, refrigerant levels, and airflow (fans running, ducts unobstructed); replace HVAC filters if clogged (should be changed every 3 months in dusty environments); verify thermal insulation of container is intact; consider HVAC capacity upgrade if system cannot maintain

Need Help Interpreting Your BESS Data?

Get a free 15-minute BESS health assessment from our UAE-based energy storage team

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